Ex. No. 6 — Inference of Q Factor of Rectangular Cavity Resonator for TE_{101} Mode with the Following Inputs: a) Dimension a in x-axis b) Dimension b in y-axis c) Dimension c in z-axis

Test Case 1

Design a rectangular cavity resonator operating in the TE_{101} mode with a = 5 cm, b = 4 cm, c = 10 cm, made of copper ($\sigma = 5.8\times10^7$ S/m). Examine how the Q-factor changes as dimension c is varied from 8 cm to 12 cm, with a = 4 cm and b = 3 cm held constant.

Scilab Program

```
clear; clc;
a = 5*10^-2; b = 4*10^-2; c = 10*10^-2;   // set a, b, c per row (meters)
C = 5.8*10^7; Uo = 4*%pi*10^-7;
f101 = 3.335*10^9;
d = sqrt(1/(%pi*f101*Uo*C));
Q = (a*a+c*c)*a*b*c/(d*(2*b*(a^3 + c^3)+a*c*(a*a+c*c)));
disp(Q, 'Quality factor of TE101 = ');
```

Result Table

a (cm)	b (cm)	c (cm)	Sim. Q Factor	Theo. Q Factor
4.0	3.0	8.0	11155.6576	11155.6576
4.0	3.0	9.0	11096.4312	11096.4312
4.0	3.0	10.0	11034.2245	11034.2245
4.0	3.0	11.0	10975.7477	10975.7477
4.0	3.0	12.0	10923.2481	10923.2481

Result: The Q-factor of the rectangular cavity resonator for the TE_{101} mode, obtained by varying the dimensions, is determined using Scilab. Increasing c slightly decreases the Q-factor here, since a larger c raises the wall surface area faster than it raises the stored volume for this a, b combination.

Test Case 2

Optimize the rectangular cavity resonator to achieve the highest possible Q-factor, based on the ranges of a, b and c explored in Test Cases 1, 3 and 4 (a: 4–6 cm, b: 3–5 cm, c: 8–12 cm).

Scilab Program

```
clear; clc;
a = 5*10^-2; b = 4*10^-2; c = 10*10^-2;   // set a, b, c per row (meters)
C = 5.8*10^7; Uo = 4*%pi*10^-7;
f101 = 3.335*10^9;
d = sqrt(1/(%pi*f101*Uo*C));
Q = (a*a+c*c)*a*b*c/(d*(2*b*(a^3 + c^3)+a*c*(a*a+c*c)));
disp(Q, 'Quality factor of TE101 = ');
```

Result Table — Highest Q-Factor Combination

a (cm)	b (cm)	c (cm)	Sim. Q Factor	Theo. Q Factor
6.0	5.0	8.0	17361.4539	17361.4539

The maximum a (6 cm), maximum b (5 cm) and minimum c (8 cm) within the explored ranges give the highest Q-factor: Q rises with a and b (more stored volume relative to the wall area facing those axes) and falls as c grows over this range, so the best combination sits at the a/b upper bound and the c lower bound.

Test Case 3

Assess how a single dimension affects resonator performance: fix b = 4 cm and c = 10 cm, and vary a between 4 cm and 6 cm in equal steps.

Scilab Program

```
clear; clc;
a = 5*10^-2; b = 4*10^-2; c = 10*10^-2;   // set a, b, c per row (meters)
C = 5.8*10^7; Uo = 4*%pi*10^-7;
f101 = 3.335*10^9;
d = sqrt(1/(%pi*f101*Uo*C));
Q = (a*a+c*c)*a*b*c/(d*(2*b*(a^3 + c^3)+a*c*(a*a+c*c)));
disp(Q, 'Quality factor of TE101 = ');
```

Q-Factor for Varying a

a (cm)	b (cm)	c (cm)	Sim. Q Factor	Theo. Q Factor
4.0	4.0	10.0	12331.8421	12331.8421
4.5	4.0	10.0	13376.4936	13376.4936
5.0	4.0	10.0	14325.5713	14325.5713
5.5	4.0	10.0	15180.8629	15180.8629
6.0	4.0	10.0	15945.2065	15945.2065

The Q-factor increases steadily as a increases from 4 cm to 6 cm, showing the resonator is fairly sensitive to a. In practice, the largest dimension generally has the greatest influence on Q, since it dominates the volume-to-surface-area ratio that Q depends on.

Test Case 4

Investigate how dimension b affects the Q-factor: fix a = 5 cm and c = 10 cm, and vary b between 3 cm and 5 cm.

Scilab Program

```
clear; clc;
a = 5*10^-2; b = 4*10^-2; c = 10*10^-2;   // set a, b, c per row (meters)
C = 5.8*10^7; Uo = 4*%pi*10^-7;
f101 = 3.335*10^9;
d = sqrt(1/(%pi*f101*Uo*C));
Q = (a*a+c*c)*a*b*c/(d*(2*b*(a^3 + c^3)+a*c*(a*a+c*c)));
disp(Q, 'Quality factor of TE101 = ');
```

Q-Factor for Varying b

a (cm)	b (cm)	c (cm)	Sim. Q Factor	Theo. Q Factor
5.0	3.0	10.0	12603.7478	12603.7478
5.0	3.5	10.0	13533.2277	13533.2277
5.0	4.0	10.0	14325.5713	14325.5713
5.0	4.5	10.0	15009.0432	15009.0432
5.0	5.0	10.0	15604.6401	15604.6401

Result

The Q-factor of a rectangular cavity resonator operating in the TE_{101} mode was determined using Scilab for a range of dimensions a, b and c. Q increases with both a and b and decreases with c over the ranges studied, and the highest Q-factor within the explored ranges was obtained at a = 6 cm, b = 5 cm, c = 8 cm.